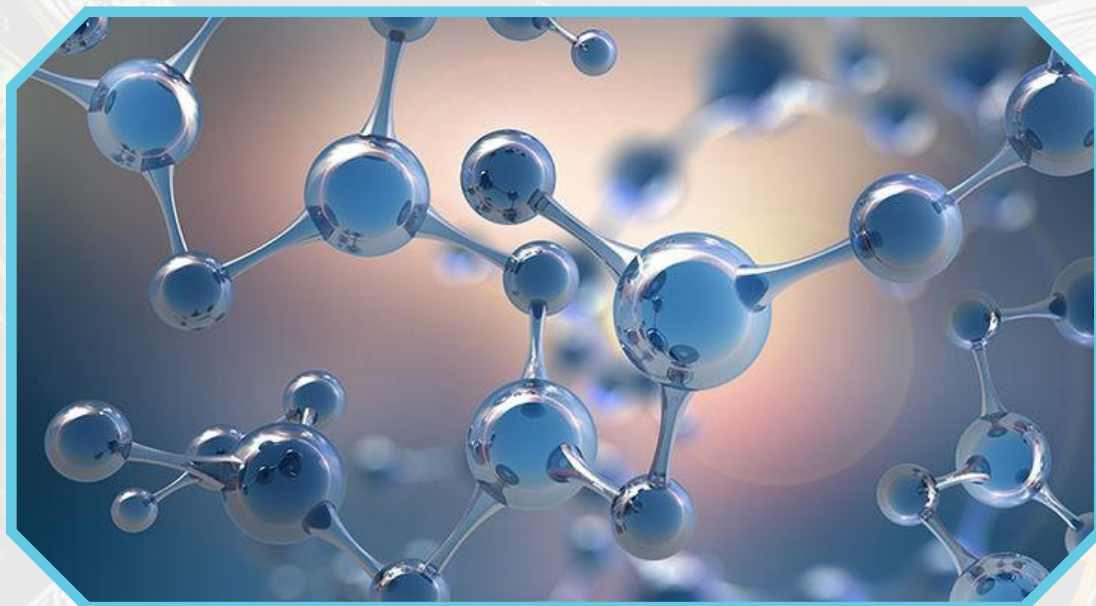




UNIT 15

MOLECULAR THEORY OF GASES



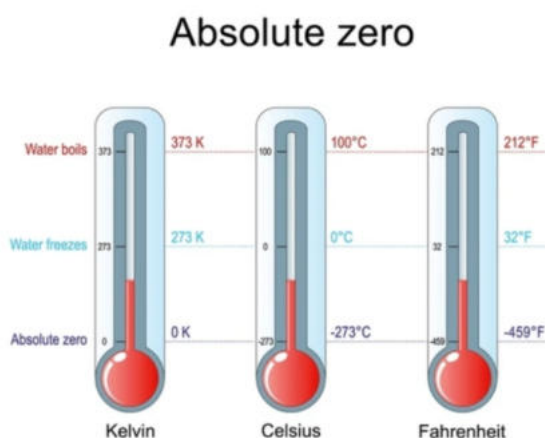
Temperature:

Temperature is a measure of the average translational kinetic energy of the molecules of a body.

Scales of Temperature:

There are three scales of temperature:

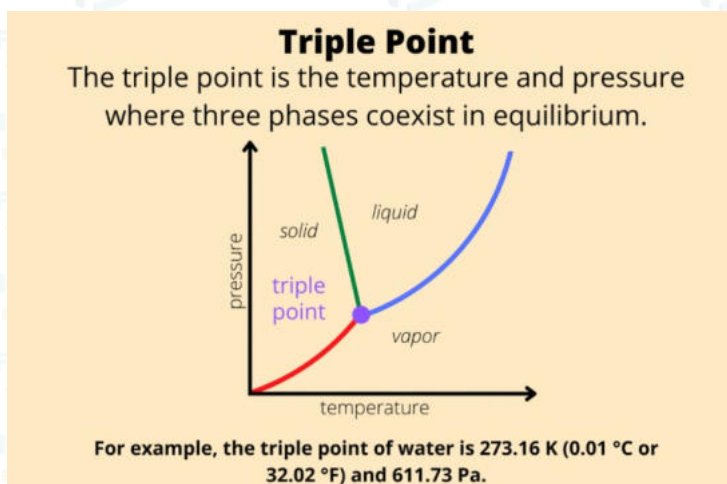
- (i) Centigrade or Celsius scale
- (ii) Fahrenheit scale
- (iii) Kelvin or absolute scale are commonly used these days



The Triple Point of Water:

The vapor, liquid, and solid states of water can coexist in thermal equilibrium only at a specific pressure and temperature.

The triple point of water occurs at a specific temperature and pressure, where all three phases are in thermodynamic equilibrium. This means that at the triple point, ice, liquid water, and water vapor can exist together without any phase changing to another.



Gas Laws

Boyle's Law:

Boyle's law states that 'Volume V of a given mass of a gas is inversely proportional to the pressure P , provided the Temperature T of the gas remains constant.'

Boyle's law can be written as:

$$V \propto \frac{1}{P} \text{ (at constant temperature)}$$

$$PV = \text{constant}$$

Boyle's law can also be written as:

$$P_1V_1 = P_2V_2 = \text{Constant}$$

Charles's Law:

This law states that 'The volume V of a given mass of a gas is directly proportional to the absolute temperature T at constant pressure P .'

Charles's law can be written as:

$$V \propto T \text{ (at constant pressure)}$$

Or

$$\frac{V}{T} = \text{constant}$$

Charles's law shows the relation between the law as:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

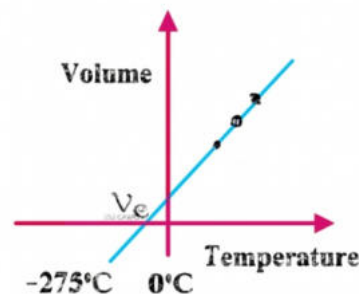
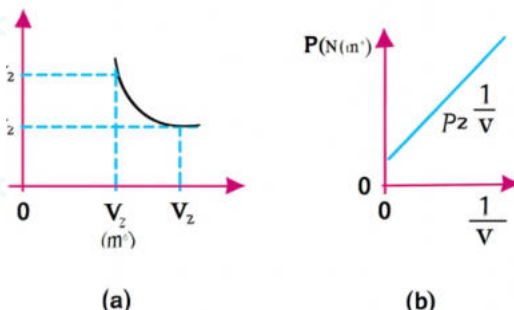
Avogadro's Law:

Avogadro's law states that 'Equal volume of all gases contains the same number of molecules at the same temperature and pressure.'

$$V \propto n \text{ (at constant temperature and pressure)}$$

General Gas Law:

In order to derive the general gas law, we make use of Boyle's law, Charles's law, and Avogadro's law. An interrelation among the physical quantities.



According to Boyle's law:

$V \propto \frac{1}{p}$ (when number of moles and temperature T are kept constant)

According to Charles's law:

$V \propto T$ (when n and pressure P are kept constant)

According to Avogadro's law:

$V \propto n$ (when T and P are kept constant)

Combined Relations:

Consider for a moment that none of the variable are to be kept constant, then all the above three relations can be joined together.

$$V \propto \frac{nt}{p}$$

$$V = \text{constant} \frac{nt}{p}$$

$$V = \frac{Rnt}{p}$$

Where **R** is constant of proportionality and is called General gas constant or universal gas constant and does not depend on the quantity of gas in the sample.

If P is measured in Nm^{-2}

V in m^3 ,

T in Kelvin,

then the volume of universal gas constant is $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

$$PV = nRT$$

Kinetic Theory of Gases:

Basic Postulates of Kinetic Theory of Gases:

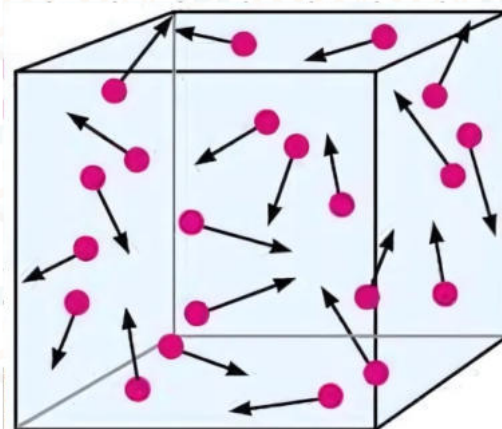
The basic postulates of Kinetic Theory of gases are as under:

- 1) A gas contains a very large number of particles called molecules. Depending on the gases each molecule consists of an atom or a group of atoms.
- 2) A finite volume of a gas consists of a very large number of molecules. This assumption is justified by experiments. At standard conditions there are 3×10^{25} molecules per cubic meter.
- 3) The size of the molecules is much smaller than separation between molecules; it is about $3 \times 10^{-10} \text{ m}$.

Pressure of Gas:

The pressure exerted by a gas is merely the momentum transferred to the walls of the container per second per unit area due to the continuous collisions of molecules of the gas.

In order to calculate the pressure of an ideal gas from Kinetic Theory, let us consider a cube having side length L whose walls are perfectly elastic, containing N number of molecules, each of mass m as shown in the figure.



Consider a single molecule of mass m moving with velocity V parallel to the x -axis. It moves back and forth, colliding at regular intervals with the ends of the box and thereby contributing to the pressure of the gas.

A molecule which has a velocity V can be resolved into three rectangular components V_x , V_y and V_z parallel to three co-ordinates axis x , y and z .

A molecule which collides with the face $ABCD$ of the cube, it will rebound elastically in opposite direction, such that x -component of the velocity V_x is reversed, while the V_y and V_z remain unaffected. Therefore, the momentum before collision is mV_x and after collision is $-mV_x$ causing a change of momentum.

Change in momentum: $P_i - P_f = mV_x - (-mV_x) = mV_x + mV_x$

Change in momentum = $2mV_x$

$$\Delta P = 2mV_x$$

After recoil, the molecule travels to opposite face and collides with it, rebounds and travels back to the face $ABCD$ after covering a distance $2L$. The time Δt between two successive collisions with face $ABCD$ is:

$$\Delta t = \frac{2L}{V_{1x}}$$

$$F_1 = \frac{\Delta p}{\Delta t}$$

$$F_1 = \frac{2mV_{1x}}{\left(\frac{2L}{V_{1x}}\right)}$$

$$F_1 = 2mV_{1x} \times \frac{V_{1x}}{2L}$$

$$F_1 = mV_{1x} \times \frac{V_{1x}}{L}$$

$$F_1 = \frac{mV_{1x}^2}{L}$$

Similarly, the forces due to all other molecules can be determined. Thus, the total X-directed force F due to N number of molecules of the gas moving with velocities $V_1, V_2, V_3, \dots, V_n$, is:

$$F = F_1 + F_2 + F_3 + \dots + F_n$$

$$F = \frac{mV_1^2x}{L} + \frac{mV_2^2x}{L} + \frac{mV_3^2x}{L} + \dots + \frac{mV_n^2x}{L}$$

$$F = \frac{m}{l}(V_1^2x + V_2^2x + V_3^2x + \dots + V_n^2x)$$

Multiply and divide by N:

$$F = \frac{mn}{l} \left(\frac{V_1^2x + V_2^2x + V_3^2x + \dots + V_n^2x}{N} \right)$$

$$V_x^2 = \left(\frac{V_1^2x + V_2^2x + V_3^2x + \dots + V_n^2x}{N} \right)$$

$$F = \frac{mNV_x^2}{L}$$

since

$$P = \frac{F}{A} \text{ and } A = L^2$$

$$P = \frac{mNV_x^2}{l \frac{L^2}{L^2}}$$

$$\text{But, } \rho = \frac{mN}{L^3} = \frac{mN}{V} \text{ (} V = \text{ volume of gas)}$$

$$P = \rho V_x^2$$

The terms V_x^2 is only one component of the total velocity.

Since $v^2 = V_x^2 + V_y^2 + V_z^2$ on the average, $V_x^2 = V_y^2 = V_z^2$ due to randomness of the molecular motion.

Substituting this value into the above equation, we find that:

$$V^2 = V_x^2 + V_y^2 + V_z^2$$

$$V^2 = 3V_x^2$$

$$V_x^2 = \frac{V^2}{3}$$

$$P = \frac{\rho V^2}{3}$$

$$P = \frac{1}{3} \rho V^2$$

The relation between Kinetic Energy of Molecules and Temperature:

The relation between molecular Kinetic Energy and temperature can be derived by using

Gas Law and Kinetic Energy Calculation:

$$P = \frac{1}{3} \rho V^2$$

$$\text{But, } \rho = \frac{mN}{L^3}$$

$$P = \frac{1}{3} \frac{mN}{V} V^2$$

$$3PV = mNV^2$$

$$\text{Here, } PV = nRT$$

$$3nRT = mNV^2$$

$$\text{But, } n = \frac{N}{N_A}$$

$$3 \frac{R}{N_A} RT = mNV^2$$

$$3 \frac{R}{N_A} T = mV^2$$

$$\text{Since, } K = \frac{R}{N_A}$$

$$3KT = mV^2$$

Multiply 1/2 on both sides

$$\frac{3}{2} KT = \frac{1}{2} mV^2$$

As we know that, K.E_{av} = $\frac{1}{2} mV^2$

$$K.E_{AV} = \frac{3}{2} KT$$

The mean translational kinetic energy of a molecule of an ideal gas is proportional to the Absolute temperature.

Hence, mean translational kinetic energy of a molecule $\propto T$.